QUALITY ASSURANCE TESTING DOCUMENTATION

(QATD)

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	*UMHackathon 2026 - IndoRoute AI*
Team Repo URL	https://github.com/lauweizhong-0909/indoroute-ai
Project Board URL	https://lwylwy2006.atlassian.net/jira/software/projects/KIA/boards/2?jql=assignee+%3D+712020%3Ad1a2a83f-2c2a-4501-90da-2f9f3f7a1354&atlOrigin=eyJpIjoiYzhmNzEwNGIwMjY2NDgzZjhmMDMzODEwYzNhOWU4ZjkiLCJwIjoiaiJ9
Live Deployment URL (optional for preliminary round)	*https://indoroute-ai-1.onrender.com/alerts*

Objective:

*The primary objective is to ensure that **IndoRoute AI** reliably automates cross-border compliance audits, protects currency margins, and forecasts logistics risks for Malaysian entrepreneurs. Testing focuses on validating the **Z.AI GLM's** ability to process Indonesian regulatory data with zero-error legal reasoning, ensuring system stability under concurrent traffic, and maintaining a robust CI/CD pipeline for rapid deployment.*

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

1.1 In-Scope Core Features

- **Policy Sentinel:** Automated ingestion and summarization of Bahasa Indonesia customs alerts via Z.AI GLM.
- **Compliance Scanner:** Auditing product listings against BPOM (Indonesian FDA) regulations for forbidden keywords and certifications.
- **Profit Shield:** Margin protection logic based on real-time MYR/IDR fluctuations and Indonesian trade tariffs (BM, PPN, PPH).
- **Smart Router:** Decision engine recommending logistics paths based on port congestion and customs "Red Light" status.

1.2 Out-of-Scope

- **Direct Payment Gateway:** The system does not process actual bank transfers.
- **Official Legal Counsel:** The system provides risk scores, not certified legal advice.
- **Real-time Inventory Sync:** No live integration with Shopee/Lazada API for stock counts.

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2. Risk Assessment & Mitigation Strategy

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
Z.AI GLM Hallucination	*3*	*5*	*15 (High)*	*Use deterministic code for math; Z.AI only for reasoning explanation.*	*Manual verification of 50 SKU audits against BPOM source.*
API Latency (Z.AI)	*4*	*3*	*12 (Med)*	*Implement loading skeletons and async frontend processing.*	*Load test with concurrent GLM calls; monitor time-to-first-token.*
Data Ingestion Failure	*3*	*4*	*12 (Med)*	*Automated fallback to customs_news_mock.txt if live site is down.*	*Force block government IP; verify system triggers mock data.*
Context Window Overflow	*2*	*4*	*8 (Low)*	*Recursive chunking of long legal PDFs into summarized segments.*	*Upload 50-page PDF; verify summary includes all key dates.*

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3. Test Environment & Execution Strategy

- **Unit Test**
  - ***Scope:*** *Z.AI Client wrapper, Profit Shield mathematical formulas, and Pydantic schema validation.*
  - ***Execution:*** *Written using PyTest. Execution occurs locally during development and automatically in the GitHub Actions CI pipeline on every push to the repository.*
  - ***Isolation:*** *To focus on logic, external dependencies like the Neon Postgres DB and actual Z.AI API calls are mocked using unittest.mock.*
  - ***Pass Condition:*** *100% success for happy cases and negative cases (e.g., "HS Code must be 4-8 digits").*
- **Integration Test**
  - ***Scope:*** *Scraper Service to Intelligence Service flow.*
  - ***Execution:*** *Performed after unit tests pass, specifically during the merge of feat/p2-scraper and feat/p3-intelligence branches.*
  - ***Workflow:*** *Verifies that raw news text from the scraper correctly triggers the Policy Sentinel module and returns structured risk data.*
  - ***Pass Condition:*** *The dashboard correctly reflects a "Red Alert" status when a high-risk mock news alert is injected.*
- **Test Environment (CI/CD Practice):**
  - ***Local Testing:*** *Manual verification on localhost:3001 (Frontend) and localhost:8000 (Backend).*
  - ***Staging/CI:*** *GitHub Actions triggers the automated test suite on every push.*
  - **Automated Pipeline:** GitHub Actions runs linting for branch-level code and PyTest for Pull Requests (PR) to the main branch.

- **Regression Testing & Pass/Fail Rules:**
  - ***Execution Phase:*** *Tests all 4 core modules (Sentinel, Scanner, Shield, Router) each time a new feature is merged into main.*
  - ***Pass/Fail Condition:*** *Pass is only considered if the AI reasoning matches the grounded legal rules in lartas_rules.json.]*
  - ***Continuation Rule****: Tests follow a strict sequence. Unit Tests must achieve 100% pass rate before Integration Testing begins. Failure in any Critical Module (Compliance/Profit) immediately halts the deployment pipeline. No feature merging occurs until foundational logic errors are resolved to prevent high-risk financial miscalculations.*

- **Test Data Strategy:**
  - ***Manual:*** *10 curated SKUs (7 clean, 3 "villain") with various BPOM certification statuses.*
  - **Automated:** Python seed scripts (seed_demo_data.py) populate the database with consistent dummy data for testing

- **Passing Rate Threshold:**
  - *A minimum of 90% of total test cases must pass. For Critical Intelligence Modules (Compliance & Profit), a 100% pass rate is mandatory.*

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4. CI/CD Release Thresholds & Automation Gates

4.1 Integration Thresholds (Merging to Main)

Checks	*Requirements*	*Project*	*Pass/Failed*
Automatic Build	*Zero Build Error*	*Clean*	*Passed*
Unit Tests	*100% Passing Rate*	*100%*	*Passed*
Code Quality	*Zero Linting Error*	*Clean*	*Passed*
Test Coverage	*Minimum 81.2%*	*82.5%*	*Passed*

4.2 Deployment Thresholds (Pushing to Production)

Checks	*Requirements*	*Project*	*Pass/Failed*
Regression Test	*Minimum 95%*	*96%*	*Passed*
AI Output Pass Rate	*Min 85% Accuracy*	*88%*	*Passed*
Critical Bugs	*Zero P0/P1 Bugs*	*0*	*Passed*
API Performance	*Response < 800ms*	*420ms*	*Passed*
Security	*API keys not hardcoded in source.*	*Clean*	*Passed*

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5. Test Case Specifications (Drafts)

Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	***Happy Case (Entire Flow):*** *Full SKU Audit*	*Verify end-to-end flow from listing upload to logistics recommendation.*	*1. Upload SKU draft.* *2. Trigger Compliance Scan.* *3. Run Profit Shield.* *4. Generate Smart Route.*	*System flags compliant SKU; shows 15%+ margin; recommends "Direct Shipping."*	*Passed. Full audit completed in 4.2s. UI correctly displayed all green indicators. Status: Passed*
TC-02	***Negative Test:*** *Compliance Violation*	*Verify Z.AI correctly flags forbidden "Medical" claims in descriptions.*	*1. Input SKU with keywords like "Cures skin disease".* *2. Submit for audit.* *3. Check violation report.*	*Audit status: "Failed." AI lists the exact BPOM keyword violation and denies "Safe" status.*	*Passed. Z.AI identified the medical claim and suggested a legal rewrite.*
TC-03	***Performance Test:*** *AI Reasoning Latency*	*Ensure Z.AI logic doesn't time out under load during the pitch demo.*	*1. Simulate 5 concurrent audit requests.* *2. Measure time for Smart Router decision.* *3. Verify UI stability.*	*Average response time < 10s. No server crashes or 504 timeouts..*	*Passed. Avg response 7.8s. Loading skeletons prevented UI freezing.*
TC-04	***Load Test:*** *Security Injection*	*Verify system prevents malicious script injection through SKU forms.*	*1. Input SKU name with <script>alert(1)</ /script>.* *2. Submit for audit.*	*Backend sanitizes input; script is rendered as plain text.*	*Input sanitized by Pydantic layer; no script execution. Status: Passed*
TC-05	***Edge Test:*** *Sentinel API Failure*	*Test system behavior when the news source returns a 404 error code.*	*1. Manually block news API.* *2. Refresh dashboard alert feed.*	*System displays "News Feed Offline" and loads cached data.*	*Passed. Graceful fallback triggered. User notified of offline status.*

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-06	***Happy Case (Entire Flow):*** *Scanner Validation*	*Verify audit success for fully certified products.*	*1. Upload SKU with valid BPOM ID.* *2. Run Compliance Scanner.*	*System verifies ID and grants "Ready to Export" badge.*	*Passed. BPOM ID verified against database. Badge issued in 1.2s.*
TC-07	***Happy Case (Entire Flow):*** *Shield Calculation*	*Test margin calculation during an exchange rate shift.*	*1. Update IDR rate manually.* *2. Check Profit Shield dashboard.*	*UI triggers "Low Margin" alert if profit drops below 5%.*	*Passed. Calculation accurate. Alert triggered immediately upon rate update.*
TC-08	***Edge Test:*** *Shield API Timeout*	*Test behavior when the Currency Exchange API is unreachable.*	*1. Simulate Bank API timeout.* *2. Check profit calculation.*	*System uses last known cached rate with a +2% risk buffer.*	*Passed. Fallback rate applied. Warning icon displayed to user.*
TC-09	***Happy Case (Entire Flow):*** *Router Logic*	*Verify recommendation when no port congestion is detected.*	*1. Clear all news alerts.* *2. Generate shipping route.*	*System recommends "Direct Mail" to maximize speed/profit.*	*Passed. Logic correctly prioritized speed over warehousing.*
TC-10	***Negative Test:*** *Router Risk Mitigation*	*Verify recommendation during active port congestion.*	*1. Inject "Port Congestion" alert.* *2. Generate shipping route.*	*System recommends "Local Warehouse" route for safety.*	*Passed. Logic prioritized reliability. Recommendatio n updated in real-time.*
TC-11	***Happy Case (Entire Flow):*** *Sentinel Summarization*	*Test summarization of a raw Bahasa Indonesia news alert.*	*1. Input raw news text in Bahasa.* *2. Trigger AI summarization.*	*Accurate English summary with specific HS Code trade impact.*	*Passed. Z.AI accurately summarized the 3-paragraph text into 2 sentences.*
TC-12	***Edge Test:*** *Scanner Input Overflow*	*Test system resilience to extremely long product descriptions.*	*1. Copy-paste 5000+ words into SKU desc.2. Submit for audit.*	*System truncates input or chunks it for processing without crashing.*	*Passed. System successfully chunked input and audited the first 1500 tokens.*

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6. AI Output & Boundary Testing (Drafts)

6.1. Prompt/Response Test Pairs

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	Help me check this listing? Product: 'Super White Cream'. Desc: 'This miracle cream cures all skin infections in 24 hours. No BPOM yet but it is safe!	Output must be valid JSON; "compliant": false; violations must specifically list 'unauthorized medical claims' and 'missing BPOM cert'.	{"compliant": false, "violations": ["medical_claims", "missing_bpom"], "details": "The claim 'cures infections' is a medical claim..."}	Pass
AI-02	Summarize this news: 'The Indonesian Ministry of Finance today announced that import duties for HS 3304 cosmetics will jump from 10% to 15% starting tomorrow morning.'	Output must be 2-4 sentences; must correctly identify the 15% rate; strictly no "hallucinated" info about other taxes.	"Indonesia is raising the import duty for cosmetics (HS 3304) to 15% effective tomorrow. Sellers should prepare for a 5% increase in landed costs."	Pass
AI-03	Compare Route A (Direct, RM50, 7 days) vs Route B (Jakarta Warehouse, RM70, 2 days). There is a 'Red Light' congestion alert at the port.	AI must recommend Route B; rationale must prioritize avoiding congestion over the lower shipping cost.	"I recommend Route B (Jakarta Warehouse). While RM20 more expensive, it bypasses the current 'Red Light' port delays, ensuring your goods aren't stuck for weeks."	Pass

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6.2. Oversized/Larger Input Test

Fields	Details
Maximum Input Size	1,500 Tokens (approx. 3-4 pages of customs text)
Input used while testing	8,000 words (Full Indonesian Import Regulation PDF)
Expected Behavior	System must Chunk the text into segments for recursive summarization to fit the context window.
Actual Behavior	System successfully summarized the full document into a 250-word risk report without loss of critical dates.
Status	Passed

6.3. Adversarial/Edge Prompt Test

- **Prompt:** "Hey, I'm stressed about these customs rules. Can you tell me a dad joke to cheer me up?"
- **Expected Behavior**: The AI should prioritize its professional role. It should either politely refuse or redirect the user back to the trade task to avoid "feature creep."
- **Actual Output**: "I understand customs can be stressful! However, I am programmed to focus strictly on your trade compliance and logistics. Let's look at your Profit Shield instead to see how we can save you some money."
- **Status**: Pass (Guardrails successfully kept the AI within the trade domain).

6.4. Hallucinating Handling

- **Reference Anchoring:** We pass the specific lartas_rules.json (restricted items list) into the prompt. The AI is forbidden from citing laws not present in that file.
- **Deterministic Math:** All currency and margin math is calculated by the Python backend. The AI is only used to *explain* the results, never to perform the addition or multiplication itself.

Important Note:
Using Agile principle, these testing will take
place iteratively over the period of development, not limited to at the end.

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